

Reader Disagreement Identifies Electrocardiograms at Increased Risk of Automated Diagnostic Error

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1 Reader disagreement during ECG interpretation may identify cases that are intrinsically
2 difficult to classify, yet conventional evaluation typically collapses the adjudication process into a
3 single final reference label. Automated ECG interpretation has achieved high diagnostic perfor-
4 mance and has also been extended beyond categorical diagnosis to estimation of clinically relevant
5 values.^{1;2} Prior work has shown limited expert agreement in difficult AI–reader discordant ECGs,
6 while uncertainty estimation offers a complementary strategy for referring lower-confidence auto-
7 mated interpretations for expert review.^{3;4} We therefore first examined whether uncertainty-based
8 referral isolates an error-prone subset of ECGs and, at full coverage, whether reader disagreement
9 is enriched among error-containing ECGs and associated with increased patient-level error risk.

10 This retrospective secondary analysis used data from the Clinical Outcomes in Digital
11 Electrocardiology (CODE) study, obtained through the Telehealth Network of Minas Gerais, a
12 public telehealth system serving primary-care facilities across Minas Gerais, Brazil. Standard
13 12-lead ECGs were recorded predominantly using TEB ECGPC (Tecnologia Eletrônica Brasileira,
14 São Paulo, Brazil) or ErgoPC 13 (Micromed Biotecnologia, Brasília, Brazil) devices.¹ CODE-
15 15%, comprising 345,779 examinations from 233,770 patients recorded between 2010 and 2016,
16 served as the model-development cohort.⁵ Independent evaluation used CODE-test, comprising
17 827 ECGs from 827 unique patients collected in 2018. Each ECG was independently interpreted by
18 2 cardiologists for first-degree atrioventricular block, right bundle branch block, left bundle branch
19 block, sinus bradycardia, atrial fibrillation, and sinus tachycardia, with disagreements adjudicated
20 by a third senior specialist.¹ The present analysis used only publicly available, de-identified data
21 and involved no recruitment, intervention, or access to identifiable patient information; the source
22 CODE study was approved by the Research Ethics Committee of the Universidade Federal de Minas
23 Gerais (protocol 49368496317.7.0000.5149).¹

24 An ensemble of 5 lightweight 1-dimensional convolutional neural networks with the same
25 architecture was trained using different random seeds, and their predicted probabilities were aver-
26 aged. Following prior work, model confidence was defined from the diagnosis-specific decision
27 threshold, and whole ECGs were referred according to their least confident diagnosis.³

Selective analysis was performed at the ECG level. For each ECG, confidence was defined by its least confident diagnosis, and progressively lower-confidence ECGs were referred according to the previously described uncertainty framework.³ Selective risk was defined as the proportion of retained ECGs containing at least 1 incorrect diagnosis among the 6 labels. Confidence intervals were obtained by bootstrap resampling of whole ECGs so that all 6 labels from each ECG remained clustered.

To characterize ECGs contributing to the error burden, reader disagreement was examined at the patient level. A prevalence ratio (PR) quantified enrichment of reader disagreement among ECGs containing any model error relative to the overall cohort. A risk ratio (RR) compared the occurrence of any model error between ECGs with reader disagreement and those with complete reader consensus. PRs, RRs, and their 95% confidence intervals were estimated from the same ECG-level bootstrap.

Uncertainty-based referral concentrated model errors in the lowest-confidence ECGs. At full coverage, 61 of 827 ECGs (7.4%) contained at least 1 diagnostic error; this decreased to 40 ECGs (5.1%) at 95% coverage and 29 ECGs (3.9%) at 90% coverage (Figure 1A). The progressive reduction in error burden as lower-confidence ECGs were referred indicates that model confidence separated more from less reliable automated interpretations.

Reader disagreement was uncommon overall but enriched among ECGs containing model error. Disagreement occurred in 33 of 827 ECGs (4.0%) overall and in 11 of 61 ECGs with any model error (18.0%; PR: 4.5; 95% CI: 2.6–6.8). Conversely, any model error occurred in 11 of 33 ECGs with reader disagreement (33.3%) compared with 50 of 794 ECGs with complete reader consensus (6.3%; RR: 5.29; 95% CI: 2.70–8.85) (Figure 1B). Thus, disagreement was disproportionately represented among error-containing ECGs and identified a small subgroup with substantially greater patient-level error risk.

In 9 of the 11 ECGs with both reader disagreement and model error, the error involved the disputed diagnosis; in the remaining 2, the model error involved a different diagnosis. This pattern suggests that disagreement usually marks difficulty in the same diagnostic decision, while

occasionally identifying a more generally difficult ECG. Importantly, disagreement should not be interpreted as evidence that the adjudicated reference was incorrect. Rather, it marks an interpretive boundary at which independent cardiologists did not initially converge, consistent with prior expert readjudication studies showing limited agreement in difficult AI–reader discordant ECGs.⁴

Model uncertainty and reader disagreement therefore captured complementary aspects of automated ECG reliability. Uncertainty provided an operational mechanism for referring lower-confidence ECGs, whereas reader disagreement was both enriched among error-containing ECGs (PR: 4.5) and associated with greater patient-level error risk (RR: 5.29). The small number of disagreement cases (33 patients, including 11 with any model error) limits precision and precludes diagnosis-specific inference; these findings should therefore be considered hypothesis-generating and require external validation in datasets preserving independent reader annotations. Nonetheless, combining uncertainty-based referral with reader-level disagreement may provide a more clinically informative assessment of automated ECG reliability than treating all adjudicated labels as equally certain.

References

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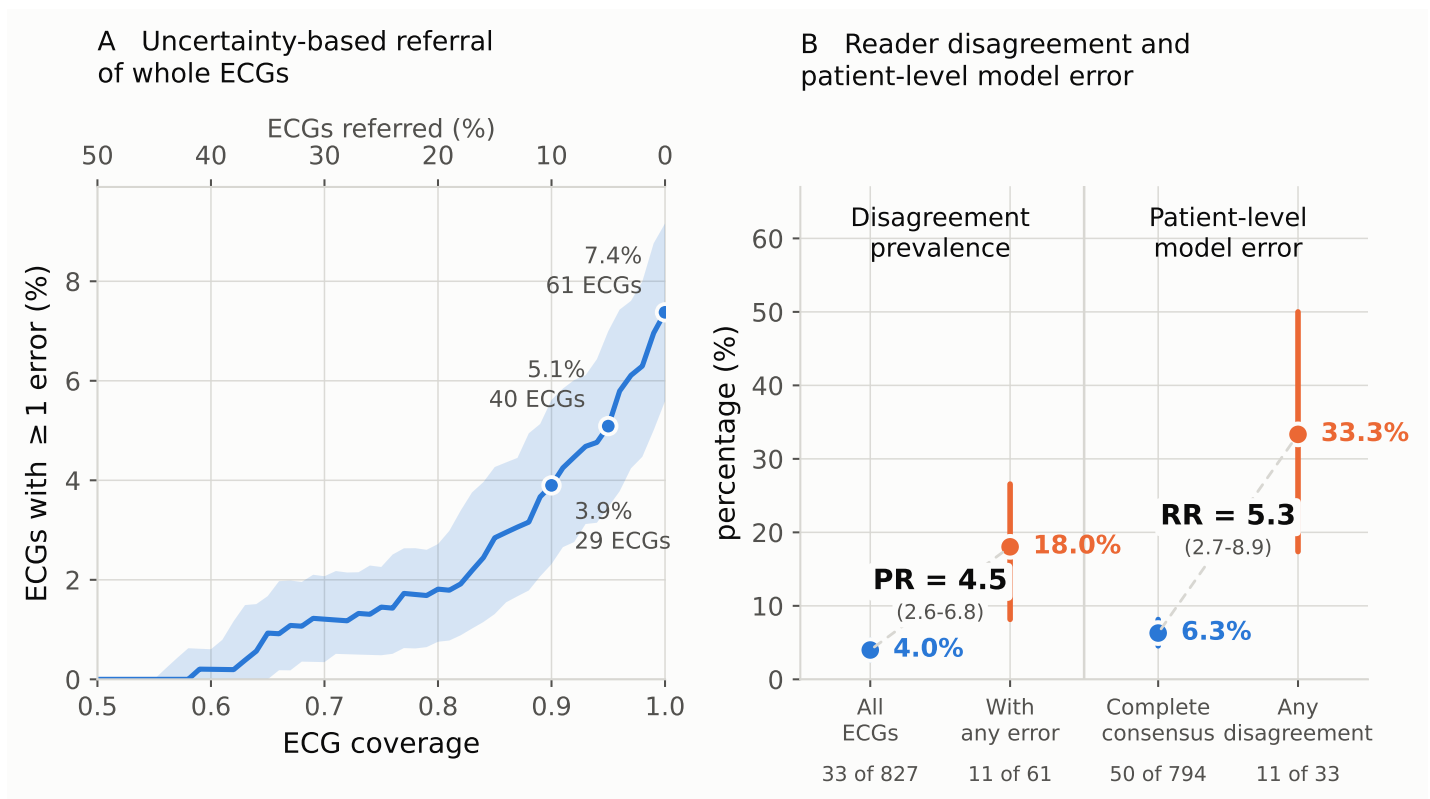


Figure 1. Uncertainty-Based Referral and Reader Disagreement in Automated ECG Diagnosis. (A) Proportion of retained ECGs containing at least 1 automated diagnostic error as lower-confidence ECGs are progressively referred; referral is by whole ECG, ranked on its least confident diagnosis, and the upper axis shows the corresponding proportion referred. Annotated points give the value and the number of affected ECGs at full, 95%, and 90% coverage. Shading represents 95% ECG-level bootstrap confidence intervals. (B) Patient-level association between reader disagreement and automated diagnostic error. Left, prevalence of reader disagreement among all ECGs and among ECGs containing at least 1 model error. Right, prevalence of any model error among ECGs with complete reader consensus and those with any disputed diagnosis. Error bars represent 95% confidence intervals. PR = prevalence ratio; RR = risk ratio.